

Introduction

This project is an e-commerce web application that simulates an online shopping platform. The system supports two user roles: administrators who manage the store inventory and view sales analytics, and customers who can browse products, manage their shopping cart, and complete purchases. The project was completed using HTML, CSS (using the Bootstrap 5 framework), JavaScript, PHP, Json files.

Features

Administrator Features

Product Management

Location: Homepage → Add product button

Administrators can add new products to the store inventory through an intuitive modal form.

The form requires:

- Product name and description
- Price and stock quantity
- Product category (existing or create new)
- Product image

The system dynamically updates the product catalog upon successful submission.

Sales Analytics Dashboard

Location: Overview

Provides comprehensive sales reporting including:

- Total profits
- Graph showing profits per product (ranked by quantity)
- Graph showing profits per product (ranked by profits)

Customer Features

Product Browsing

Location: Homepage

Customers can browse products organized by dynamically-generated categories. The interface includes:

- Category filtering system
- Product cards displaying key information
- Real-time stock availability indicators

Shopping Cart Management

Location: Homepage → Add to Cart button

Customers can browse products organized by dynamically-generated categories. The interface includes:

- Add products to cart with animated feedback
- Stock validation (prevents adding out-of-stock items)
- Quantity adjustment
- Item removal capability

Cart Review

Location: Your Cart page

Customers can:

- View all cart items with pricing details
- See total calculations
- Select specific items for checkout
- Remove unwanted items

Checkout Process

Location: Your Cart → Proceed to pay button

Simulated payment processing that:

- Validates selected items
- Processes payment instantly (simulation mode)
- Updates inventory counts
- Removes purchased items from cart

Technical Requirements

	Requirement	How it was fulfilled
1	You must work on your project on your own (no group projects)	This project was completed individually.
2	it must be a full web app, with html, css, javascript, php. Focusing only on the html and css would not suffice.	Utilizes HTML, CSS, JavaScript, and PHP across all pages
3	it must work with files, preferably json-formatted ones. You may NOT use databases for your project.	All data persisted in JSON format; no database usage
4	it must have a coherent theme, not be just a collection of technically competent pages	Unified e-commerce platform with consistent design language
5	it must have users (with Register and Login menu options, and an Admin site available only to the system administrator).	Login/register functionality with role-based access (admin vs. customer)
6	it must store data for each user (could be files the user uploaded, comments the user has made, personal information about the user, etc) and each user must be able to manage/edit their data in some way.	Upon registration, users are able to input their personal information, which is editable in the profile page. They will also be able to add items to their cart. All this information is stored in user.json.
7	it must be dynamic: pages must change via interactivity with the user or when changes happen on the server.	Real-time UI updates based on user interactions. An example of this includes how the page changes when the user clicks on different product categories to view in the homepage.
8	it must have parts that are constructed dynamically without hardwired limitations, such as any number of rows in a table (not a hardwired value), any number of images (not just a hardwired number) etc.	Category system and product listings generated programmatically from JSON data
9	it must include a feature (e.g from W3schools) or optional topic in Web programming that you researched on your own, understood it and integrated it.	Bootstrap framework for responsive design and UI components.
10	it must include some original code in javascript and php, or code that was adapted constructively. You must	Custom PHP and JavaScript with proper attribution for adapted code

	acknowledge all code that you borrowed or adapted from other sources.	
11	it must have some kind of Javascript graphics (besides included images). Optionally, if this is what attracts you the most, it can primarily focus on dynamic javascript graphics, e.g. implementing some demo code in javascript (similar to the "moving points" assignment and the Game of Life assignment; or, for example, an adaptation of a demo "game" available on W3schools demo site, etc.).	Animated "add to cart" effect.
12	it must have some interactive features (buttons, forms, etc.) which implement some local communication (on the client, in javascript, e.g. for changing the way something is visualized in the browser) and others that implement communication with the server (php, e.g. those that send to or request some info from the server).	An example includes the log in or register button that submits a form to the corresponding php.
13	clean and well documented code, both in terms of organization of files (separate folders for css, js, php, organized in possibly several files with functions that are logically grouped together) and of code (separate functions, working on parameters that are passed to them and using local variables; well-motivated use of global variables; suggestive names for variables and functions and reasonable amount of comments when the good names are insufficient to describe what a specific function or piece of code is doing, etc..)	Code is organized into corresponding folders, descriptive names for methods and variables were used.

References

Item	Link
Bootstrap nav bar	https://getbootstrap.com/docs/4.0/components/navbar/
Bootstrap modal	https://getbootstrap.com/docs/4.0/components/modal/
Bootstrap card (product card)	https://getbootstrap.com/docs/5.0/components/card/
Using __DIR__	https://stackoverflow.com/questions/32537477/how-to-use-dir
Using _GET	https://www.w3schools.com/php/php_superglobals_get.asp
Bootstrap template for cart.html.php	https://mdbootstrap.com/docs/standard/extended/shopping-carts/
Pos of element relative to viewport	https://www.w3schools.com/jsref/met_element_getboundingclientrect.asp
Bootstrap warning	https://getbootstrap.com/docs/5.0/components/alerts/
chart.js	https://www.chartjs.org/docs/latest/getting-started/installation.html
uasort	https://www.w3schools.com/php/func_array_uasort.asp
then	https://www.w3schools.com/jsref/jsref_promise_then.asp
fetch	https://www.w3schools.com/js/js_api_fetch.asp

Future Enhancements

Potential improvements for future iterations:

- Order history and tracking system
- Product search and advanced filtering
- User reviews and ratings
- More detailed admin dashboard